

Problem

The current flowing through an element is

$$i = \begin{cases} 4 \text{ A}, & 0 < t < 1 \\ 4t^2 \text{ A}, & t > 1 \end{cases}$$

Calculate the charge entering the element from $t = 0$ to $t = 2$ s.

Step-by-step solution

Step 1 of 1

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$$\begin{aligned} Q &= \int_{t=0}^2 i \, dt \\ &= \int_{t=0}^1 4 \, dt + \int_{t=1}^2 4t^2 \, dt \\ &= 4t \Big|_0^1 + \frac{4}{3} t^3 \Big|_1^2 \\ &= 4 + \frac{4}{3}(8-1) \\ &= 13.333 \text{ C} \end{aligned}$$

Therefore, the charge entering the element from $t = 0$ to $t = 2$ s is 13.333 C.